

GUSTAVO LUJAN CENTENO

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EDUCATION

University of Massachusetts Lowell

Lowell MA

Bachelor of Science in Computer Engineering

May 2026

Accelerated Master of Science in Computer Engineering (4+1 Program)

May 2027

SKILLS

Programming: C#, C++, Python, MATLAB, GO, ARM Assembly

Electronics: Circuit Design, Multisim, LTSpice, Oscilloscope, Waveform Generator, Breadboard

Tools: Git, Analog Discovery, Visual Studio Code, Arduino IDE, Raspberry Pi, Soldering

PROJECTS

Fabrication Machine for Microelectrode Arrays - *UML Project*

Spring 2026

- Contributed to the development of an automated system for microelectrode array fabrication as part of a multidisciplinary engineering team.
- Worked on hardware-software integration, motion control, and testing of embedded subsystems.
- Performed iterative testing, calibration, and debugging to improve system performance and automation.

Cooperative Real-Time Scheduler - *UML Project*

Fall 2025

- Implemented a cooperative real-time task scheduler in C/C++ on an AVR microcontroller using the Arduino Uno board.
- Designed scheduling logic, timing counters, and task execution control for multiple embedded tasks.
- Implemented CPU load measurement and timing management to monitor system performance under different scenarios.

Voice Message Recorder & Player – *UML Project*

Spring 2025

- Designed and implemented a recording and playback system in ARM Assembly.
- Implemented audio capture with microphone input and playback via PWM output integrated with digital and analog I/O.
- Partnered with teammates on programming, circuit design, and troubleshooting

Wind-Activated Visual Display - *UML Project*

Spring 2025

- Designed a sound-activated LED system using LM358 operational amplifier stages
- Integrated buzzer alerts that triggered under high-intensity sound (wind)
- Worked independently on the entire project in the design and benchmarks.

CERTIFICATIONS

IBM – Machine Learning for Data Science Certificate

WORK EXPERIENCE

Restaurant server | Torito Cantina Restaurant, Shrewsbury MA

June 2023 - May 2025

- Collaborated with other servers to divide large parties and ensure all tables received timely service
- Communicated with kitchen staff to coordinate orders and handle special requests for customers
- Trained new team members on restaurant procedures, improving overall team efficiency

COVID-19 Volunteer | Framingham Town, Framingham MA

June 2021 - Nov 2021

- Worked with a volunteer team to set up distribution stations for masks and sanitizers
- Rotated responsibilities with teammates including crowd guidance, handing out supplies, restocking materials and helping keep distribution events organized.
- Partnered with local health staff and other volunteers to reach vulnerable groups